OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

8.2 Inertial coordinate systems

The inertial system is a right-handed coordinate system according to ISO 8855 [6] with the axes pointing to the following directions (see Figure 11):

- x ⇒ right
- y ⇒ up
- z ⇒ coming out of drawing plane

For geographic reference, the following convention applies:

- x ⇒ east
- y ⇒ north
- z ⇒ up

Elements like objects and signals can be placed within the inertial coordinate system by applying a heading, followed by pitch, followed by roll:

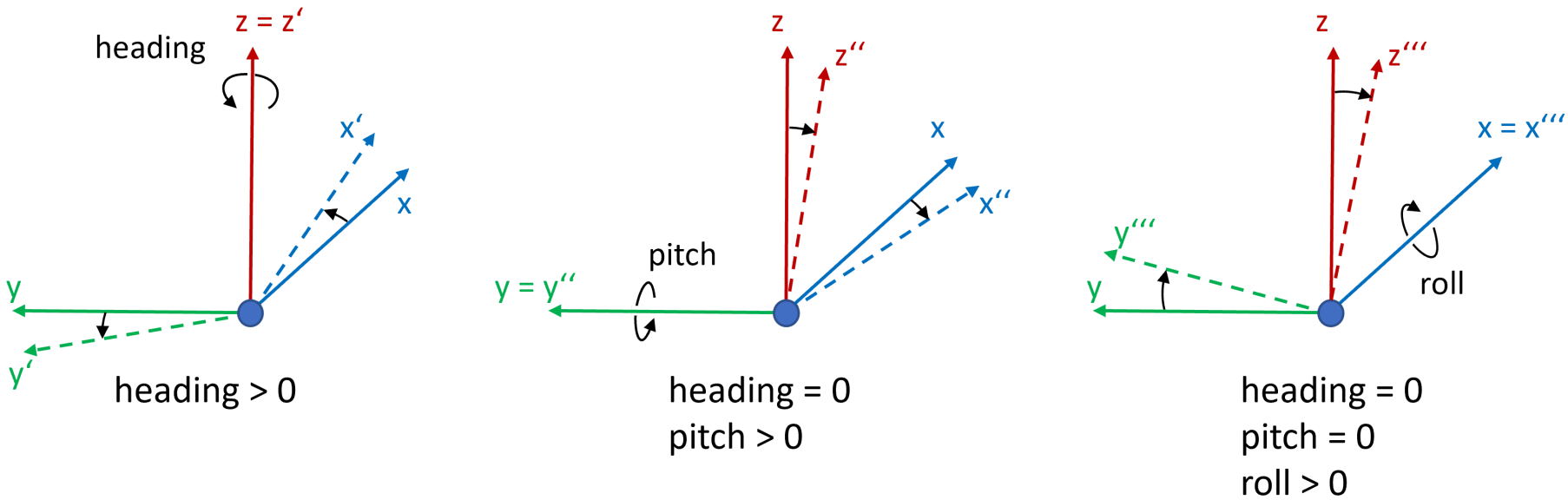

Figure 11. Inertial coordinate system with defined rotations

Figure 11 shows the positive axes and positive directions of the corresponding angles.

heading	around z-axis, where
heading = 0.0:	x' points into direction of x-axis / east
heading = +π/2:	x' points into direction of y-axis / north
pitch	around y'-axis, where
pitch = 0.0:	x"/y" plane = x'/y' plane
pitch = +π/2:	direction x" = - z' = -z
roll	around x"-axis, where
roll = 0.0:	x'''/y''' plane = x"/y" plane
roll = +π/2:	direction z''' = - y"

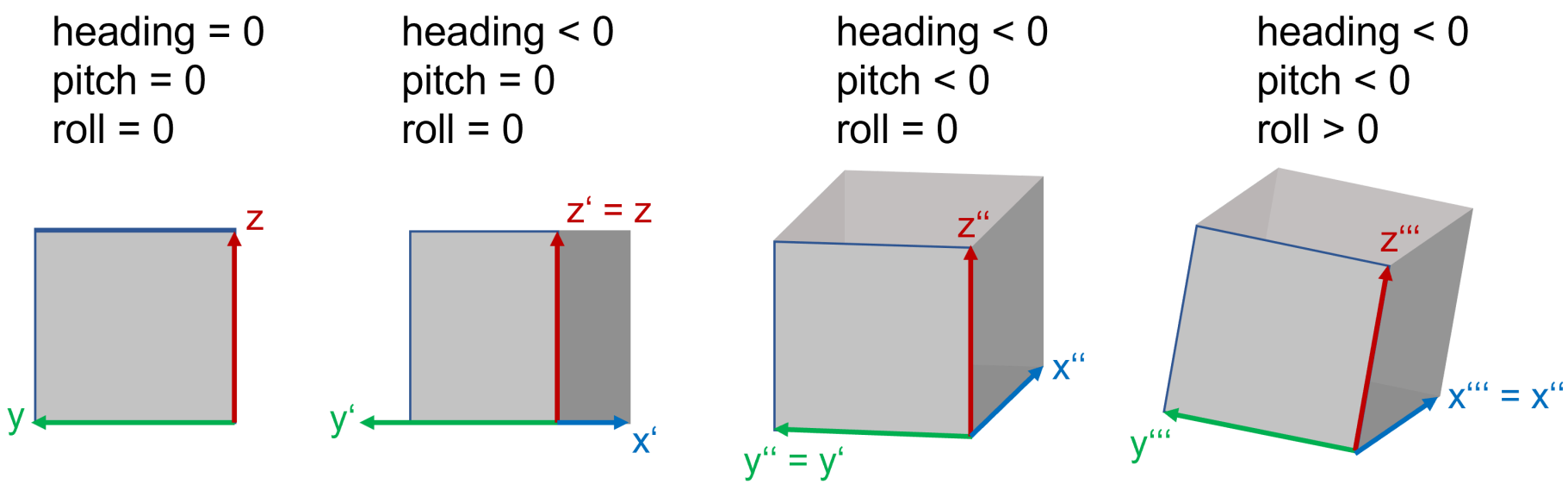

Figure 12. Inertial coordinate system with defined rotations

Figure 12 shows the different states of an inertial coordinate system with defined rotations. x’/y’/(z’=z) denotes the coordinate system after rotating x/y/z with the heading angle around the z-axis. The coordinate system x”/(y”=y’)/z” denotes the coordinate system after rotating x’/y’/z’ with the pitch angle around the y’-axis. The final rotated coordinate system (x”’=x”)/y”’/z”’ is obtained after rotating system x”/y”/z” with roll angle.